

COP2073C Programming Assignment 2

Concept Review: Element Extraction and Vector Operations

Element extraction with vector operations allows you to perform calculations on entire columns or rows of a matrix at once, without needing to manually iterate through each element.

Example: Suppose you have a matrix `sales` where the first column represents sales figures from Year 1 and the second column represents sales figures from Year 2. You can calculate the percentage increase in sales for each row using vector operations like this:

```
perc_incr <- (sales[,2] - sales[,1]) / sales[,1]
```

This approach automatically applies the calculation to all rows in the matrix without needing to manually iterate over each row individually.

Instructions

The 10-year population increases for four states that start with the letter 'A' between the years 2000 and 2010 are shown below:

State	2000	2010
Alabama	4447100	4779735
Alaska	626932	710231
Arizona	5130632	6329013
Arkansas	2673400	2915921

Write an R script that performs the following tasks:

- Create two vectors representing the populations for the years 2000 and 2010. Then, use `cbind` to create a 4x2 population matrix from these two vectors.
- Using element extraction with vector operations** create a vector that contains the percentage increase in population for each state between 2000 and 2010.
 - Hint:** Use this formula: $(2010 \text{ value} - 2000 \text{ value}) / 2000 \text{ value}$.
- Use the percentage increase vector from step b and **continuing to use element extraction with vector operations**, calculate a new vector that predicts the populations for the year 2020.
 - Hint:** Use this formula: $2010 \text{ value} * (1 + \text{percentage increase value})$.
- Use `cbind` to add the estimated 2020 population vector to the population matrix. Overwrite the original matrix object with this new one.

Expected results:

Initial population vectors for 2000 and 2010:

4447100 626932 5130632 2673400
4779735 710231 6329013 2915921

Initial population matrix:

	a2000	a2010
[1,]	4447100	4779735
[2,]	626932	710231
[3,]	5130632	6329013
[4,]	2673400	2915921

Percentage increase vector (remember that you must calculate this using element extraction, do not use constant values!)

[1] 0.07479818 0.13286768 0.23357376 0.09071632

Estimated 2020 population vector:

[1] 5137250.5 804597.7 7807304.4 3180442.6

Final population matrix:

	a2000	a2010	a2020
[1,]	4447100	4779735	5137250.5
[2,]	626932	710231	804597.7
[3,]	5130632	6329013	7807304.4
[4,]	2673400	2915921	3180442.6